T.C.
IZMIR DEMOCRACY UNIVERSITY
FACULTY OF ENGINEERING

GRADUATION PROJECT
ENGINEERING
DEPARTMENT

A Comparative Study of GAN-Based Image-to-Image Translation Methods for AFM-to-O2A Microscopy Images

by
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COMMITMENT

I hereby declare that this thesis has been written at the Faculty of Engineering at Izmir Democracy University in accordance with academical and ethical rules and all the literature informations that is used in this thesis mentioned references.

Tahir KURTAR

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LIST OF ICONS AND ABBREVIATIONS

AFM	Atomic Force Microscopy
O2A	Second Harmonic s-SNOM Amplitude Image (O_2A)
GAN	Generative Adversarial Network
cGAN	Conditional Generative Adversarial Network
Pix2Pix	Image-to-image translation model based on GAN
CycleGAN	Unpaired image-to-image translation framework based on cycle-consistent GAN
GauGAN	Spatially-adaptive normalization–based GAN architecture
ESRGAN	Enhanced Super-Resolution Generative Adversarial Network
Vanilla GAN	Original GAN architecture proposed by Goodfellow et al.
U-Net	U-shaped Convolutional Neural Network Architecture
PatchGAN	Patch-based discriminator architecture
G	Generator
D	Discriminator
L1 Loss	Mean Absolute Error Loss
GAN Loss	Adversarial Loss
SSIM	Structural Similarity Index Measure
PSNR	Peak Signal-to-Noise Ratio

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İzmir, 2025

ABSTRACT

Graduation Project

A Comparative Study of GAN-Based Image-to-Image Translation Methods for AFM-to-O2A Microscopy Images

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Department of Electrical and Electronics Engineering

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This thesis investigates deep learning–based image-to-image translation methods for converting Atomic Force Microscopy (AFM) images into Second Harmonic scattering-type Scanning Near-Field Optical Microscopy (O_2A) amplitude maps. A comparative analysis of multiple Generative Adversarial Network (GAN) architectures is conducted to evaluate their effectiveness in modeling AFM-to-O_2A modality translation.

The study initially adopts the standard Pix2Pix framework utilizing a U-Net generator and a PatchGAN discriminator to establish a strong supervised baseline trained on paired AFM–O_2A samples using a combination of adversarial and L_1 pixel-wise loss functions. Building upon this foundation, additional GAN-based architectures, including CycleGAN for unpaired translation, ESRGAN enhanced pipelines for super-resolution refinement, GauGAN for spatially-aware synthesis, and a Vanilla GAN baseline, are systematically implemented and evaluated.

Experimental results on a held-out test set indicate that supervised Pix2Pix-based models provide stable and competitive quantitative performance in terms of L_1 error, Structural Similarity Index (SSIM), and Peak Signal-to-Noise Ratio (PSNR), while alternative GAN architectures exhibit distinct trade-offs between numerical accuracy and visual realism. Notably, GauGAN achieves the strongest perceptual quality and spatial coherence in the synthesized O_2A maps, despite only marginal differences in quantitative metrics. These findings extend prior Pix2Pix-based studies [1] by offering a unified comparative evaluation of GAN architectures for nanoscale microscopy and demonstrate the feasibility of synthesizing O_2A amplitude maps directly from AFM data.

Keywords: AFM-to-O_2A translation, GAN comparison, Pix2Pix, CycleGAN, ESRGAN, GauGAN, Vanilla GAN, image-to-image translation, SSIM, PSNR

2025, 11 pages

ÖZET

Bitirme Projesi

Derin Öğrenme Tabanlı Pix2Pix Çerçevesi Kullanılarak AFM-O2A Görüntü Çevirisi

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Bu tez, Atomik Kuvvet Mikroskobu (AFM) görüntülerinin İkinci Harmonik saçılmalı Yakın Alan Optik Mikroskobu (O_2A) genlik haritalarına dönüştürülmesi amacıyla derin öğrenme tabanlı görüntüden-görüntüye çeviri yöntemlerini incelemektedir. AFM-den O_2A'ya modalite dönüşümünü modelleme etkinliklerini değerlendirmek üzere birden fazla Üretici Karşıt Ağ (GAN) mimarisi üzerinde karşılaştırmalı bir analiz gerçekleştirilmiştir.

Çalışmada ilk olarak, eşleşmiş AFM–O_2A örnekleri üzerinde denetimli bir temel model oluşturmak amacıyla, U-Net tabanlı bir üretici ve PatchGAN ayırt edici kullanan standart Pix2Pix çerçevesi benimsenmiştir. Model, karşıt kayıp ve L_1 piksel-bazlı kayıp fonksiyonlarının birleşimiyle eğitilmiştir. Bu temel yapı üzerine inşa edilerek, eşleşmemiş veriyle dönüşüm için CycleGAN, süper çözünürlük iyileştirmesi için ESRGAN destekli yaklaşımlar, mekânsal farkındalıklı sentez için GauGAN ve bir Vanilla GAN temel modeli olmak üzere ek GAN tabanlı mimariler sistematik olarak uygulanmış ve değerlendirilmiştir.

Ayrı bir test kümesi üzerinde elde edilen deneysel sonuçlar, denetimli Pix2Pix tabanlı modellerin L_1 hata, Yapısal Benzerlik İndeksi (SSIM) ve Tepe Sinyal-Gürültü Oranı (PSNR) açısından kararlı ve rekabetçi nicel performans sunduğunu göstermektedir. Buna karşılık, diğer GAN mimarileri sayısal doğruluk ile görsel gerçekçilik arasında farklı ödünleşimler sergilemektedir. Özellikle GauGAN, nicel metriklerde yalnızca sınırlı farklar olmasına rağmen, sentezlenen O_2A haritalarında en güçlü algısal kaliteyi ve mekânsal tutarlılığı sağlamaktadır. Bu bulgular, önceki Pix2Pix tabanlı çalışmaları [1] nanoskopik mikroskopi çevirimi için GAN mimarilerinin bütüncül bir karşılaştırmasıyla genişletmekte ve AFM verilerinden doğrudan sentetik O_2A genlik haritaları üretmenin uygulanabilirliğini ortaya koymaktadır.

Anahtar Kelimeler: AFM-den O_2A'ya dönüşüm, GAN karşılaştırması, Pix2Pix, CycleGAN, ESRGAN, GauGAN, Vanilla GAN, görüntüden-görüntüye çeviri, SSIM, PSNR

2025, 11 sayfa

1. INTRODUCTION

Image-to-image translation, which aims to transform images from one domain to another while preserving underlying structural content, has become a prominent research area in computer vision and deep learning. Since the introduction of Generative Adversarial Networks (GANs) by Goodfellow et al. (2014), substantial progress has been achieved in applications such as style transfer, image enhancement, and cross-modality translation [2]. In particular, conditional GANs (cGANs), most notably the Pix2Pix framework proposed by Isola et al. (2017), have demonstrated strong performance in supervised, paired-image translation tasks including maps-to-satellite imagery and edge-to-photo synthesis [3].

In the field of nanoscale microscopy, cross-modality translation has recently gained attention in overcoming hardware limitations and improving accessibility to advanced imaging techniques. Atomic Force Microscopy (AFM) provides high-resolution surface topography, whereas Second Harmonic scattering-type Scanning Near-Field Optical Microscopy (O_2A) amplitude maps deliver complementary optical contrast related to material properties at the nanoscale. However, s-SNOM systems required for O_2A imaging are expensive, complex, and not widely available. Recent studies, such as the work by Karsenty et al. (2025), have shown that deep learning models can infer s-SNOM-related information from AFM data, indicating the feasibility of synthetic optical imaging. Nevertheless, existing approaches are largely limited to a single model configuration and lack a systematic comparison of different GAN-based image-to-image translation strategies for AFM-to-O_2A conversion.

This thesis addresses this gap by conducting a comparative study of multiple GAN architectures for AFM-to-O_2A image translation. A Pix2Pix-based cGAN employing a U-Net generator and PatchGAN discriminator is first established as a supervised baseline using paired AFM–O_2A datasets and a combination of adversarial and L_1 pixel-wise loss functions. Building upon this baseline, additional GAN-based models—including CycleGAN for unpaired translation, ESRGAN-enhanced pipelines for super-resolution refinement, GauGAN for spatially-aware synthesis, and a Vanilla GAN baseline—are implemented and evaluated. Model performance is assessed quantitatively using L_1 error, Structural Similarity Index (SSIM), and Peak Signal-to-Noise Ratio (PSNR), as well as qualitatively through visual inspection of synthesized O_2A maps.

Through this comparative analysis, the study aims to identify the strengths and limitations of different GAN architectures in nanoscale modality translation and to demonstrate the feasibility of generating high-fidelity synthetic O_2A images from standard AFM inputs. By reducing reliance on specialized optical instrumentation, this work contributes toward broader access to nanoscale optical information and provides practical guidance for future deep learning–based microscopy applications.

2. GENERAL INFORMATION / RELEVANT LITERATURE

Image-to-image translation has become a prominent application of deep learning, particularly for cross-modality transformations in scientific and microscopic imaging. Conditional Generative Adversarial Networks (cGANs) enable domain-specific image synthesis by jointly optimizing adversarial objectives and pixel-level constraints, such as L_1 or perceptual loss functions, thereby preserving structural consistency between input and output domains [4]. Among these approaches, the Pix2Pix framework introduced by Isola et al. (2017) employs a U-Net–based generator and a PatchGAN discriminator and has demonstrated strong performance in supervised, paired-image translation tasks, including maps-to-satellite imagery and sketches-to-photographs.

Beyond Pix2Pix, several GAN-based architectures have been proposed to address different limitations in image-to-image translation. CycleGAN extends GAN-based translation to unpaired datasets by enforcing cycle-consistency constraints, enabling learning in scenarios where paired data are unavailable. ESRGAN focuses on perceptual quality and super-resolution refinement through residual-in-residual dense blocks and adversarial training, while GauGAN incorporates semantic and spatial conditioning to generate photorealistic images guided by layout information. Vanilla GAN models, although simpler, are often used as baseline references to highlight the benefits of more structured architectures. Together, these models represent distinct design philosophies with varying trade-offs between numerical accuracy, visual realism, and training stability.

In the context of nanoscale microscopy, deep learning–based image synthesis has gained increasing attention as a means of reducing reliance on expensive and specialized imaging hardware. Physics-informed approaches, such as the work by Chen and Dal Negro (2021), have demonstrated that neural networks constrained by physical laws can recover complex optical parameters from near-field measurements, including electric permittivity distributions derived from Maxwell's equations [5]. More recently, Stanciu et al. (2025) employed a Pix2Pix-style conditional GAN to infer scattering-type scanning near-field optical microscopy (s-SNOM) amplitude maps from Atomic Force Microscopy (AFM) images, showing that optical contrast information can be accurately synthesized from topographical inputs alone [6].

While these studies validate the feasibility of AFM-to-O_2A image translation, existing work predominantly focuses on a single model configuration and lacks a systematic comparison of alternative GAN architectures. Consequently, the relative strengths and limitations of supervised, unpaired, and enhancement-focused GAN models in this nanoscale translation task remain insufficiently explored. Motivated by this gap, the present thesis conducts a structured comparative analysis of multiple GAN-based image-to-image translation methods for AFM-to-O_2A conversion. The evaluated models are assessed quantitatively using L_1 error, Structural Similarity Index (SSIM), and Peak Signal-to-Noise Ratio (PSNR), as well as qualitatively through visual inspection, with the overarching goal of identifying robust and scalable deep learning strategies for synthetic nanoscale optical imaging.

3. EXPERIMENTAL PART

3.1 Dataset Description

The dataset used in this study consists of paired Atomic Force Microscopy (AFM) images and corresponding Second Harmonic scattering-type Scanning Near-Field Optical Microscopy (O_2A) amplitude maps acquired from the same nanoscale regions. AFM images provide high-resolution topographical information, while O_2A maps represent optical amplitude contrast obtained via s-SNOM measurements. All image pairs are spatially aligned and resized to a fixed resolution to ensure pixel-wise correspondence required for supervised learning.

The dataset used in this study consists of paired Atomic Force Microscopy (AFM) images and corresponding Second Harmonic scattering-type Scanning Near-Field Optical Microscopy (O_2A) amplitude maps acquired from the same nanoscale regions. AFM images provide high-resolution topographical information, while O_2A maps represent optical amplitude contrast obtained via s-SNOM measurements. All image pairs are spatially aligned and resized to a fixed resolution to ensure pixel-wise correspondence required for supervised learning.

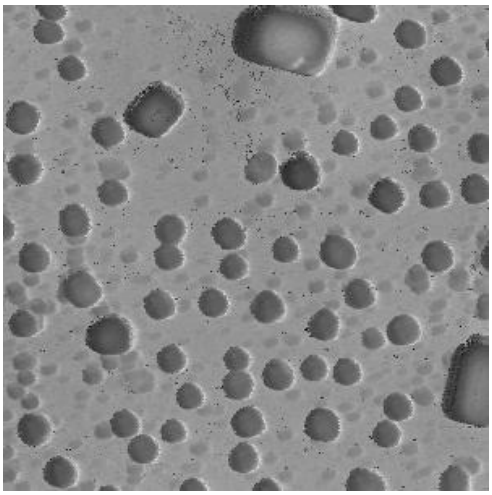


Figure 1: Second Harmonic scattering-type Scanning Near-Field Optical Microscopy (O_2A)

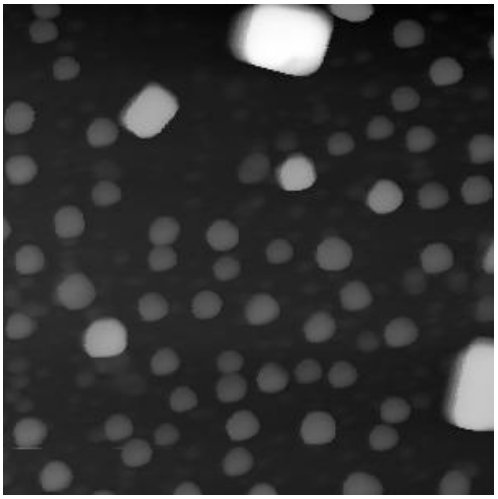


Figure 2: Atomic Force Microscopy (AFM)

3.2 GAN Architectures

To evaluate the effectiveness of different image-to-image translation strategies, multiple GAN-based architectures are implemented and compared.

3.2.1 Pix2Pix (Supervised Baseline)

The Pix2Pix model is employed as the primary supervised baseline. The generator follows a U-Net architecture with encoder–decoder structure and skip connections that preserve spatial information across scales. The discriminator is implemented as a PatchGAN, which classifies local image patches as real or fake, encouraging high-frequency detail synthesis. The model is trained on paired AFM–O_2A images using a combination of adversarial loss and L_1 pixel-wise reconstruction loss.

3.2.2 CycleGAN (Unpaired Translation)

CycleGAN is implemented to evaluate unpaired image-to-image translation performance. The architecture consists of two generators and two discriminators with cycle-consistency loss enforcing bidirectional mapping between AFM and O_2A domains. Although paired data are available, CycleGAN serves to assess the impact of removing strict pixel-level supervision on translation quality.

3.2.3 ESRGAN-Enhanced Pipeline

To investigate the effect of super-resolution refinement, an ESRGAN-based enhancement module is integrated into the translation pipeline. In this approach, the initially generated O_2A images are further refined using ESRGAN to improve perceptual sharpness and high-frequency details. This pipeline emphasizes visual quality over strict pixel-wise accuracy.

3.2.4 GauGAN (Spatially-Aware Synthesis)

GauGAN is adapted to explore spatially-aware image synthesis by incorporating structural information from AFM images. The model leverages semantic and spatial conditioning to guide the generation process, enabling the assessment of layout-driven synthesis capabilities in nanoscale microscopy translation.

3.2.5 Vanilla GAN (Baseline Reference)

A vanilla GAN model is implemented as a minimal baseline to highlight the benefits of more structured GAN architectures. This model consists of a simple convolutional generator and discriminator trained solely with adversarial loss, without explicit reconstruction constraints.

3.3 Training Configuration

All models are trained using the Adam optimizer with fixed learning rates and momentum parameters. Training is performed for a fixed number of epochs with early stopping based on validation loss to prevent overfitting. Batch size, learning rate, and loss weight parameters are kept consistent across models where applicable to ensure a fair comparison.
Model training and evaluation are conducted using the PyTorch deep learning framework on GPU-enabled hardware. Random seeds are fixed for reproducibility.

3.4 Evaluation Metrics

The performance of each model is evaluated quantitatively on the held-out test set using L_1 loss, Structural Similarity Index (SSIM), and Peak Signal-to-Noise Ratio (PSNR). L_1 loss measures pixel-wise reconstruction accuracy, while SSIM and PSNR assess perceptual similarity and signal fidelity, respectively. In addition, qualitative evaluation is performed through visual inspection of generated O_2A maps to analyze texture realism, edge preservation, and overall structural consistency.

$$\mathrm{PSNR} = 10 \times \lg\left(\frac{255^2}{\mathrm{MSE}}\right)$$

$$\mathrm{MSE} = \frac{1}{M \times N}\sum_{i=1}^{N}\sum_{j=1}^{M}\left[I(i,j) - I'(i,j)\right]^2$$

Figure 3: PSNR Formula

$$\mathrm{SSIM}(\mathbf{x},\mathbf{y}) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)\,(\sigma_x^2 + \sigma_y^2 + C_2)}.$$

Figure 4: SSIM Formula

3.5 Implementation Details

All experiments are implemented in Python using PyTorch. Model checkpoints, training logs, and evaluation results are systematically recorded to ensure traceability. Visualization of intermediate and final outputs is performed to support qualitative comparisons between different GAN architectures.

4. RESEARCH FINDINGS AND DISCUSSIONS

This section presents a quantitative and qualitative comparison of different GAN-based image-to-image translation models applied to AFM-to-O_2A microscopy image synthesis. The evaluated models include Pix2Pix, Pix2Pix enhanced with ESRGAN, GauGAN, Vanilla GAN, and CycleGAN (A→B and B→A). Performance is assessed using standard image quality metrics as well as visual inspection of the generated O_2A amplitude maps.

4.1 Quantitative Evaluation

The quantitative performance of all models is summarized in Table 1, where L_1 loss, Structural Similarity Index (SSIM), and Peak Signal-to-Noise Ratio (PSNR) are reported.

	Pix2Pix	Pix2Pix + ESRGAN	GauGAN	Vanilla GAN	Cycle GAN A--->B	Cycle GAN B--->A
L1 Loss	0.2084	0.2062	0.202	0.2054	1.8326	1.8326
SSIM	0.4422	0.4419	0.457	0.4255	0.2421	0.2216
PSNR	19.48 dB	19.39 dB	19.49 dB	19.22 dB	12.86 dB	11.04 dB

Table 1: Quantitative comparison of GAN-based models

As shown in Table 1, Pix2Pix, Pix2Pix + ESRGAN, GauGAN, and Vanilla GAN exhibit relatively close numerical performance across all metrics. Among these, **GauGAN achieves the lowest L_1 loss (0.202) and the highest SSIM (0.457)**, indicating improved structural consistency with the ground-truth O_2A images. Pix2Pix slightly outperforms Vanilla GAN in both SSIM and PSNR, confirming the benefit of conditional supervision using paired data.

The Pix2Pix + ESRGAN pipeline shows marginal improvement in L_1 loss compared to the baseline Pix2Pix; however, the gains remain limited in terms of SSIM and PSNR. This suggests that ESRGAN primarily enhances perceptual sharpness and noise suppression rather than learning new physically meaningful features from AFM inputs.

In contrast, CycleGAN demonstrates significantly poorer quantitative performance in both A→B and B→A directions, with substantially higher L_1 loss and lower SSIM and PSNR values. This behavior is expected due to the unpaired training setup, which lacks strong pixel-wise constraints necessary for accurate AFM-to-O_2A mapping.

4.2 Qualitative Visual Comparison

To complement the quantitative analysis, Figure 3 presents representative examples of generated O_2A images alongside the corresponding ground truth.

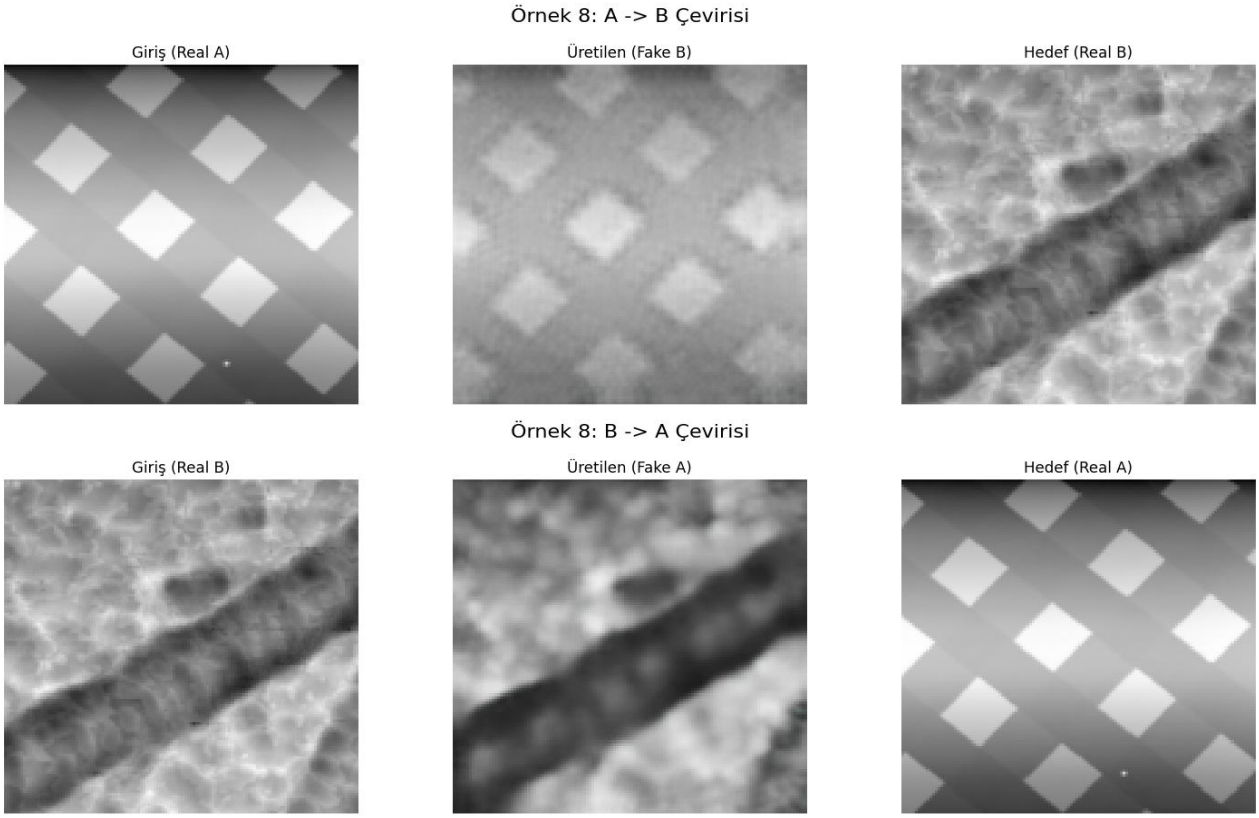
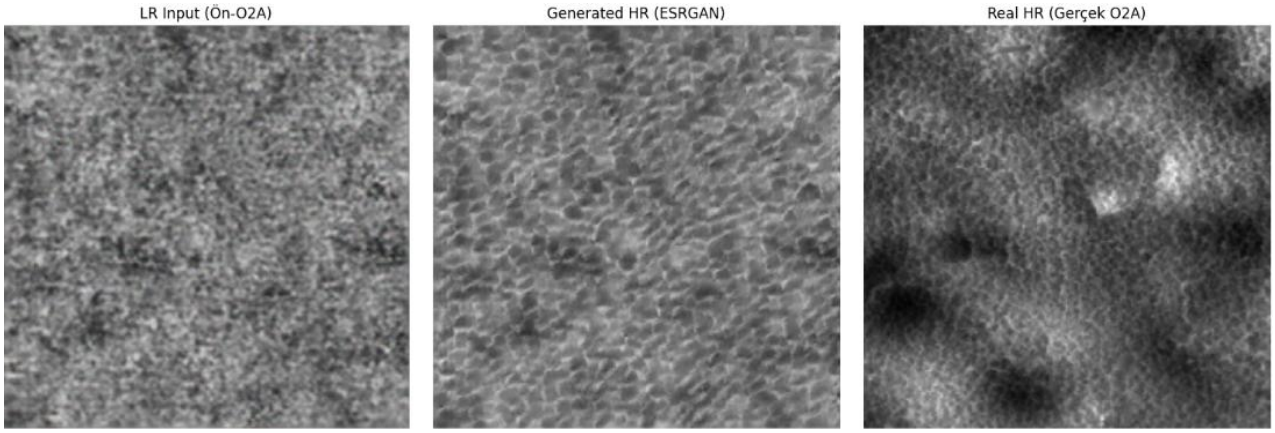


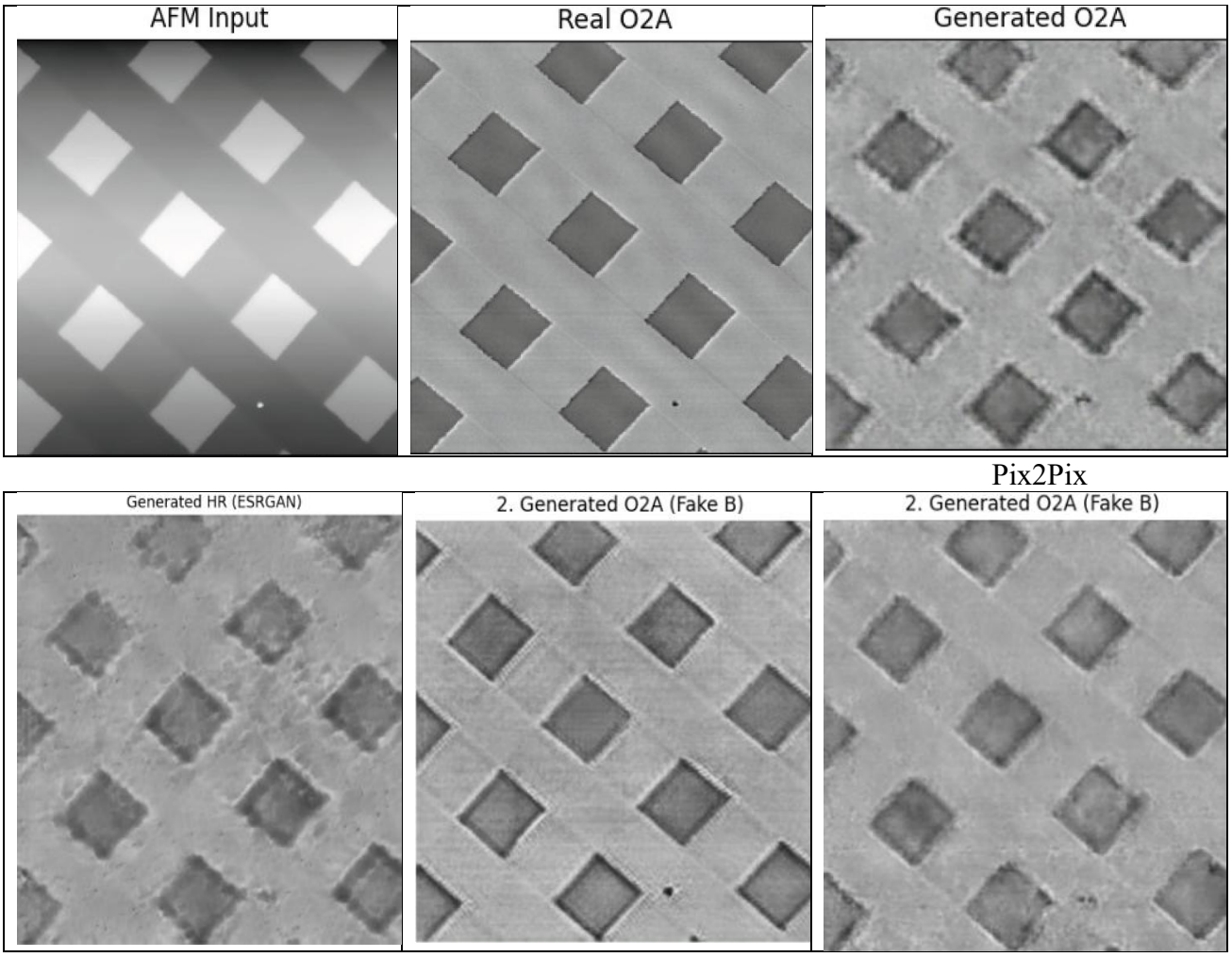
Figure 5: Visual comparison of generated O_2A images for Cycle GAN



Pix2Pix Output ESRGAN Real O2A

Figure 6: ESRGAN-enhanced O_2A image synthesis results

ESRGAN enhances sharpness without improving structural fidelity.



Pix2Pix

Pix2Pix + ESRGAN GauGAN Vanilla GAN

Figure 7: Visual comparison of generated O_2A images for different models

Visual inspection reveals clear differences in image quality that are not fully captured by numerical metrics alone. GauGAN-generated images appear sharper and more spatially coherent, with improved edge definition and reduced blurring in nanoscale features. Although Pix2Pix and Vanilla GAN produce visually similar outputs, Pix2Pix preserves structural details slightly more consistently, aligning with its superior SSIM and PSNR values.

The Pix2Pix + ESRGAN outputs exhibit smoother textures and reduced high-frequency noise; however, this improvement is largely cosmetic and does not significantly enhance feature fidelity. In the case of CycleGAN, the generated images lack clear correspondence to true O_2A contrast, confirming that unpaired translation is less suitable for this task.

4.3 Discussion of Model Behavior

Overall, the results indicate that paired, conditional GAN frameworks are more effective for AFM-to-O_2A image translation. While numerical differences between Pix2Pix-based methods are relatively small, qualitative evaluation highlights GauGAN's advantage in producing visually sharper and more interpretable images. This emphasizes the importance of combining quantitative metrics with visual analysis when evaluating microscopy image synthesis models.

5. CONCLUSIONS, CONTRIBUTIONS AND FUTURE WORK

This study presented a comparative evaluation of multiple GAN-based image-to-image translation architectures for converting Atomic Force Microscopy (AFM) images into Second Harmonic scattering-type Scanning Near-Field Optical Microscopy (O_2A) amplitude maps. Supervised, unpaired, and enhancement-oriented GAN models were assessed under a unified experimental framework to analyze their quantitative accuracy and qualitative visual fidelity in nanoscale modality translation.

Quantitative results indicate that **GauGAN achieves the lowest L_1 loss and the highest SSIM and PSNR values**, demonstrating strong structural preservation and signal fidelity. The supervised Pix2Pix model also exhibits competitive and stable performance across all metrics, confirming its effectiveness as a reliable baseline when paired AFM–O_2A data are available. The Pix2Pix + ESRGAN pipeline provides marginal numerical improvements primarily through noise suppression and visual smoothing rather than enhanced feature learning. In contrast, CycleGAN models trained on unpaired data show significantly lower performance in both A→B and B→A directions, reflecting the inherent limitations of unpaired translation for precise nanoscale reconstruction.

Qualitative visual comparison further reveals that **GauGAN produces the sharpest and most visually coherent O_2A images**, better preserving fine spatial details compared to other architectures. These findings highlight an important trade-off between numerical metrics and perceptual image quality, indicating that models optimized for spatial realism may outperform traditional supervised approaches in visual interpretation tasks.

Overall, the results confirm the feasibility of synthesizing O_2A amplitude maps from standard AFM data using GAN-based methods. This comparative study extends prior Pix2Pix-focused work by demonstrating that spatially-aware architectures such as GauGAN can offer superior visual quality while maintaining strong quantitative performance. Future work will focus on bridging perceptual and numerical optimization through physics-informed constraints, uncertainty estimation, and hybrid loss functions to further improve robustness and interpretability in synthetic optical microscopy.

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RESUME

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